Supplemental Material for: A Conditional Landauer–Nelson Bridge with Non-Markovian TICE Closure

Kevin Henry Miller*
Founder and President, Q-Bond Network DeSCI DAO, LLC and
(Dated: March 18, 2026)

I. S1. FULL DERIVATION CHAIN

A. S1.1. Caldeira–Leggett: bath spectral density to noise kernel

Start from the Ohmic–Drude spectral density:

$$J(\omega) = m\gamma_0\omega\,\frac{\Omega^2}{\omega^2+\Omega^2}. \tag{1}$$

The symmetrized noise kernel at temperature T is:

$$N(\tau) = \frac{1}{\pi}\int_0^\infty J(\omega)\coth\!\Big(\frac{\hbar\omega}{2k_{\mathrm{B}}T}\Big)\cos(\omega\tau)\,d\omega. \tag{2}$$

In the high-T limit $\hbar\omega \ll k_{\mathrm{B}}T$ for all ω in the support of (**??**):

$$\coth\!\Big(\frac{\hbar\omega}{2k_{\mathrm{B}}T}\Big) \approx \frac{2k_{\mathrm{B}}T}{\hbar\omega}, \tag{3}$$

so that

$$N(\tau) \simeq \frac{2m\gamma_0 k_{\mathrm{B}}T\Omega^2}{\pi\hbar}\int_0^\infty \frac{\cos(\omega\tau)}{\omega^2+\Omega^2}\,d\omega = \frac{m\gamma_0 k_{\mathrm{B}}T\Omega}{\hbar}\,e^{-\Omega|\tau|}. \tag{4}$$

B. S1.2. KMS condition to fluctuation–dissipation relation

The KMS (Kubo–Martin–Schwinger) condition for the bath correlation function at inverse temperature $\beta = 1/(k_{\mathrm{B}}T)$ gives:

$$\tilde{N}(\omega) = \coth\!\Big(\frac{\hbar\omega}{2k_{\mathrm{B}}T}\Big)\,\tilde{D}(\omega), \tag{5}$$

where $\tilde{D}(\omega) = J(\omega)/\pi$. In the high-$T$ Markov limit this reduces to:

$$N(\tau) = \frac{k_{\mathrm{B}}T}{\hbar}\,\eta(\tau), \tag{6}$$

where $\eta(\tau) = m\gamma_0\,\Omega\,e^{-\Omega|\tau|}$ is the dissipation kernel.

C. S1.3. Green–Kubo: friction to thermal diffusion

From the generalized Langevin equation the Green–Kubo relation gives:

$$D_{\mathrm{th}} = \frac{k_{\mathrm{B}}T}{m\gamma_0}. \tag{7}$$

D. S1.4. Nelson diffusion

In Nelson's stochastic mechanics, the diffusion coefficient associated with the forward and backward Wiener processes is:

$$D_{\mathrm{q}} = \frac{\hbar}{2m}. \tag{8}$$

E. S1.5. Closure to theorem

Under Conjecture 1 ($D_{\mathrm{th}} = 2D_{\mathrm{q}}$):

$$\frac{k_{\mathrm{B}}T}{m\gamma_0} = 2\cdot\frac{\hbar}{2m} = \frac{\hbar}{m} \tag{9}$$

$$\Rightarrow \quad \gamma_0 = \frac{k_{\mathrm{B}}T}{\hbar}. \tag{10}$$

Then $R = D_{\mathrm{th}}/D_{\mathrm{q}} = 2$ and $E_{\mathrm{bridge}} = k_{\mathrm{B}}T\ln 2 = E_{\mathrm{Landauer}}$.

II. S2. THREE KERNEL SOLUTIONS AND CROSSING TIMES

We give the effective friction, TICE closure factor, and crossing time t^* for three physically motivated kernels.

A. S2.1. Ohmic–Drude (exponential) kernel

$K(\tau) = A\,e^{-\Omega|\tau|}$, with $A = mk_{\mathrm{B}}T\gamma_0\Omega/\hbar$.
Effective friction integrated against $|\psi|^2 = 1$ (uniform

* Kevin@qbondnetwork.com; https://www.qbondnetwork.com

profile):

$$\gamma_{\rm eff}(t) = \gamma_0 + \frac{\beta A}{\Omega}\big(1 - e^{-\Omega t}\big), \tag{11}$$

$$C_{\rm TICE}(t) = \frac{k_{\rm B}T/\hbar}{\gamma_0 + (\beta A/\Omega)(1 - e^{-\Omega t})}, \tag{12}$$

$$t^*_{\rm Drude} = \frac{1}{\Omega}\ln\!\left(\frac{\beta A/\Omega}{\gamma_0 + \beta A/\Omega - k_{\rm B}T/\hbar}\right). \tag{13}$$

Existence condition. t^* is real and positive iff $\gamma_0 < k_{\rm B}T/\hbar < \gamma_0 + \beta A/\Omega$.

B. S2.2. Algebraic (power-law) kernel

$K(\tau) = A\,(1 + \Omega|\tau|)^{-n}$ with $n > 1$.
Integrated friction:

$$\gamma_{\rm eff}(t) = \gamma_0 + \frac{\beta A}{\Omega(n-1)}\Big[1 - (1 + \Omega t)^{1-n}\Big]. \tag{14}$$

Asymptote: $\gamma_{\rm eff}(\infty) = \gamma_0 + \beta A/[\Omega(n-1)]$.
Crossing time (implicit):

$$(1 + \Omega t^*)^{n-1} = \frac{\beta A/[\Omega(n-1)]}{\gamma_0 + \beta A/[\Omega(n-1)] - k_{\rm B}T/\hbar}. \tag{15}$$

C. S2.3. Gaussian kernel

$K(\tau) = A\,e^{-\Omega^2\tau^2/2}$.
Integrated friction:

$$\gamma_{\rm eff}(t) = \gamma_0 + \beta A\sqrt{\frac{\pi}{2\Omega^2}}\,{\rm erf}\Big(\frac{\Omega t}{\sqrt{2}}\Big). \tag{16}$$

Asymptote: $\gamma_{\rm eff}(\infty) = \gamma_0 + \beta A\sqrt{\pi/(2\Omega^2)}$.
Crossing time (implicit):

$${\rm erf}\Big(\frac{\Omega t^*}{\sqrt{2}}\Big) = \frac{k_{\rm B}T/\hbar - \gamma_0}{\beta A\sqrt{\pi/(2\Omega^2)}}. \tag{17}$$

III. S3. PERTURBATIVE EXPANSION OF DEVIATION LAW

Let $\gamma_0 = \gamma^*(1 + \delta)$ with $\gamma^* = k_{\rm B}T/\hbar$. The deviation from Landauer is:

$$\Delta E = E_{\rm bridge} - E_{\rm Landauer} = -k_{\rm B}T\ln(1 + \delta). \tag{18}$$

Taylor expansion for $|\delta| \ll 1$:

$$\Delta E = -k_{\rm B}T\Big[\delta - \frac{\delta^2}{2} + \frac{\delta^3}{3} - \cdots\Big]. \tag{19}$$

For the TICE time-dependent case with $C_{\rm TICE}(t) = 1 + \epsilon(t)$:

$$\Delta E(t) = k_{\rm B}T\ln(1 + \epsilon) \approx k_{\rm B}T\Big[\epsilon - \frac{\epsilon^2}{2} + \cdots\Big]. \tag{20}$$

The sign convention: $\epsilon > 0 \Rightarrow C_{\rm TICE} > 1 \Rightarrow$ super-Landauer (reversible work extraction possible). $\epsilon < 0 \Rightarrow$ sub-Landauer (second-law violation for irreversible erasure).

IV. S4. NELSON/TICE INTERNAL CLOSURE RATIO

Define the Nelson–TICE internal ratio:

$$\mathcal{R}(t) \equiv \frac{\text{thermal diffusion at time } t}{\text{Nelson diffusion}} = \frac{k_{\rm B}T}{m\,\gamma_{\rm eff}(t)}\cdot\frac{2m}{\hbar} = 2\,C_{\rm TICE}(t). \tag{21}$$

Timeline:

- $t = 0$: $\mathcal{R}(0) = 2k_{\rm B}T/(\hbar\gamma_0)$. If $\gamma_0 > k_{\rm B}T/\hbar$ (overdamped initial condition) then $\mathcal{R}(0) < 2$.
- $t = t^*$ (crossing): $\mathcal{R}(t^*) = 2$, exactly Landauer.
- $t \to \infty$: $\mathcal{R}(\infty) = 2k_{\rm B}T/[\hbar\,\gamma_{\rm eff}(\infty)]$.

The Landauer crossing is a *transient event* in the TICE framework, occurring when the kernel-accumulated friction drives $\gamma_{\rm eff}$ through $k_{\rm B}T/\hbar$.

V. S5. BOCHNER POSITIVITY: WHAT IT DOES AND DOES NOT PROVE

Bochner's theorem states that a function $K : \mathbb{R} \to \mathbb{C}$ is positive-definite (i.e., a valid bath correlation function) if and only if it is the Fourier transform of a non-negative measure.

What Bochner guarantees: All three kernels in Sec. **??** (Drude, algebraic, Gaussian) are positive-definite for $A, \Omega > 0$ and hence are admissible bath correlators.

What Bochner does not guarantee: Admissibility does not imply *uniqueness*. Infinitely many positive-definite kernels exist. The closure $D_{\rm th} = 2D_{\rm q}$ selects a relationship between γ_0 and T but does not select a unique kernel. The kernel shape determines how $\gamma_{\rm eff}(t)$ approaches the closure, but the asymptotic closure ratio is the same for all kernels once the integral of K is fixed.

VI. S6. NUMERICAL VERIFICATION SUMMARY

All results were verified numerically by the Q-Bond Research Engine (15 audit scripts, 388 checks, 0 failures). BT#11 specifically was tested in scripts `bt11_verification.py` and `bt19_verification.py` with the following checks:

Test	Expected	Status
$R = 2$ at $\gamma_0 = k_\mathrm{B}T/\hbar$	2.000	PASS
$E_\mathrm{bridge}/(k_\mathrm{B}T) = \ln 2$	$0.6931\ldots$	PASS
$\gamma_0\hbar/(k_\mathrm{B}T) = 1$	1.000	PASS
$\tau_\mathrm{cor} = \hbar/(k_\mathrm{B}T)$	exact match	PASS
$\partial R/\partial T = 0$	0.000	PASS
Deviation: $\delta = 0.1$	$\Delta E/k_\mathrm{B}T = -0.0953$	PASS
Deviation: $\delta = -0.1$	$\Delta E/k_\mathrm{B}T = +0.1054$	PASS
Sign rule: $\delta > 0 \Rightarrow \Delta E < 0$	correct	PASS
Sign rule: $\delta < 0 \Rightarrow \Delta E > 0$	correct	PASS
Mass independence of R	verified 5 masses	PASS
Ohmic kernel crossing	$t^* > 0$, real	PASS
$C_\mathrm{TICE}(t^*) = 1$ at crossing	1.000	PASS
Gaussian kernel crossing	$t^* > 0$, real	PASS
Algebraic kernel crossing	$t^* > 0$, real	PASS

All numerical checks use double precision with tolerance 10^{-12}.

VII. S7. CONSISTENCY WITH KNOWN RESULTS

A. S7.1. Caldeira–Leggett limit

In the Markov limit ($\Omega \to \infty$), the memory kernel $K(\tau) \to 2A\,\delta(\tau)/\Omega$ and $\gamma_\mathrm{eff} \to \gamma_0$ instantaneously. The TICE closure factor reduces to its Markov value $C_\mathrm{TICE} = k_\mathrm{B}T/(\hbar\gamma_0)$, recovering the main theorem.

B. S7.2. Fluctuation–dissipation theorem

At $C_\mathrm{TICE} = 1$, the relation $\gamma_\mathrm{eff} = k_\mathrm{B}T/\hbar$ implies $D_\mathrm{th} = k_\mathrm{B}T/(m\gamma_\mathrm{eff}) = \hbar/m = 2D_\mathrm{q}$, confirming self-consistency with the FDT at the closure point.

C. S7.3. Landauer bound

The second law requires $E_\mathrm{bridge} \geq E_\mathrm{Landauer}$ for irreversible one-bit erasure. The deviation formula gives $E_\mathrm{bridge} - E_\mathrm{Landauer} = -k_\mathrm{B}T\ln(1+\delta) \geq 0$ iff $\delta \leq 0$ (underdamped), consistent with the physical requirement that reaching Landauer requires approaching the closure from below.